St. John College of Engineering and Management
Autonomous Institute
(A Christian Religious Minority Institution)
Approved by AICTE and DTE, Affiliated to University of Mumbai / MSBTE
DTE Code : 3218 AICTE Permanent ID : 1-4790201
NAAC Accredited with Grade 'A+', Three Programs NBA Accredited

25+

End Semester Examination **SET-A**

Course Name - Engineering Physics **Semester- I**

Course Code- 25FEBS102 **Time- 10:00- 11:00 am**

Maximum Marks- 30

24 NOV 2025

All Question are Compulsory

Assume Suitable data wherever required

Q	Attempt all Questions	Marks Max	Course Outcome	RBT Level
Q1.a	Derive Bragg's law.	5	1	L2
	or			
Q1.b	Differentiate between crystalline solid and amorphous solid. Find the interplanar spacing between the family of planes (111) in crystal of lattice constant 3 Å.	3 2	1 1	L2
Q2.a	With the help of diagram, explain the variation of fermi level with temperature in n-type semiconductor.	5	2	L2
	or			
Q2.b	Find the resistivity of intrinsic Ge at 300°K given the density of carriers as 2.5×10^{19} m^{-3}. The mobility of electrons and holes is 0.39 m^2/V-sec and 0.19 m^2/V-sec. What is direct band gap semiconductor?	3 2	2 2	L2
Q3.a	Derive one-dimensional time dependent Schrodinger equation for matter waves.	5	3	L2
	Or			
Q3.b	State de-Broglie Hypothesis of matter waves and derive its different forms. Electrons accelerated through 100 V are reflected from a crystal. What is the glancing angle at which the first reflection occurs? Lattice spacing= 2.15 Å.	3 2	3 3	L2
Q4.a	Derive the conditions for maxima and minima due to interference of light reflected from thin film of uniform thickness.	5	4	L3
	or			
Q4.b	Interference fringes are produced by monochromatic light falling normally on a wedge shaped film of cellophane whose R.I is 1.4. The angle of wedge is 20 sec of arc and the distance between successive fringes is 0.25 cm. Calculate the wavelength of light. What do you mean by thin film?	3 2	4 4	L3

Q5.a	What is holography? Explain the construction and reconstruction of hologram.	5	5	L3
	or			
Q5.b	Write a short note on Spontaneous emission. Calculate the numerical aperture of a fibre with core index 1.56 and cladding index 1.50.	3 2	5 5	L3
Q6.a	Draw the schematic diagram of scanning electron microscope and explain its construction and working.	5	6	L1
	or			
Q6.b	Give a comparison of scanning electron microscope an atomic force microscope. Define nanotechnology. What are the different types of nanomaterials?	3 2	6 6	L1

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